

Factor Markets

Based on Ch. 18, Economics 10th ed., Parkin, Powell & Mathews

Course: Eco 101 (Introduction to Microeconomics)

Factor Markets

The four factors of production are:

- Labor
- Capital
- Land (natural resources)
- Entrepreneurship

Markets for Labor Services

- **Labor services** are the physical and mental work efforts people provide to produce goods and services.
- A **labor market** consists of workers who supply labor and firms that demand labor.
- The **wage rate** is the price paid for labor services.
- Most labor markets are **competitive**, with many buyers (firms) and sellers (workers).
- In competitive labor markets, the **wage rate is determined by supply and demand**, just like prices in other competitive markets.

Demand for a Factor of Production

The demand for a factor of production, such as labor, is a **derived demand**—it comes from the demand for the goods and services that labor produces.

- A firm's demand for labor depends on its **profit-maximizing level of output**.
- Firms hire the quantity of labor that produces this profit-maximizing output.

To decide how much labor to hire, a firm compares:

- the **cost of hiring an additional worker** (the wage rate), and
- the **value of the worker to the firm**, called the **value of marginal product (VMP)**.

The **value of marginal product** equals the **price of output × the marginal product of labor**.

Although labor is used as the example, these principles apply to the **demand for all factors of production**.

Value of Marginal Product

TABLE 18.1 Value of Marginal Product at Angelo's Bakery

	Quantity of labor (<i>L</i>) (workers)	Total product (<i>TP</i>) (loaves per hour)	Marginal product ($MP = \Delta TP / \Delta L$) (loaves per worker)	Value of marginal product ($VMP = MP \times P$) (dollars per worker)
A	0	0	7	14
B	1	7	 6	12
C	2	13	5	10
D	3	18	4	8
E	4	22	3	6
F	5	25		

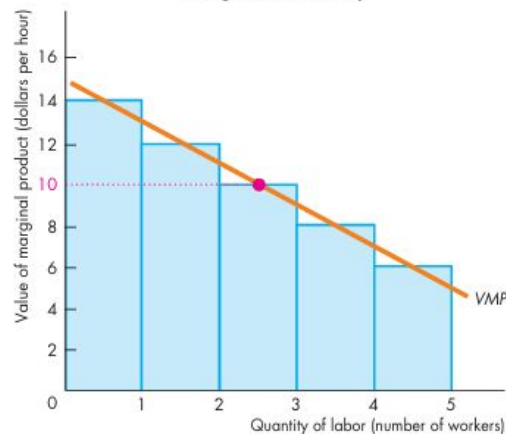
The value of the marginal product of labor equals the price of the product multiplied by marginal product of labor. If Angelo's hires 2 workers, the marginal product of the second worker is 6 loaves (in the third column). The price of a loaf is \$2, so the value of the marginal product of the second worker is \$2 a loaf multiplied by 6 loaves, which is \$12 (in fourth column).

A Firm's Demand for Labor

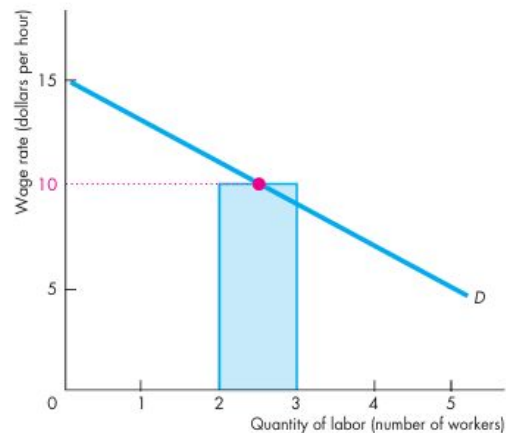
The quantity of labor demanded by a firm is the quantity at which the value of the marginal product of labor **equals** the wage rate.

A change in the wage rate brings a change in the quantity of labor demanded and a movement along the demand for labor curve.

FIGURE 18.1 The Demand for Labor at Angelo's Bakery



(a) Value of marginal product



(b) Demand for labor

Changes in a Firm's Demand for Labor

A firm's demand for labor depends on:

1. Price of the Firm's Output

- When the price of what a firm sells goes up, workers become more valuable.
- The firm hires **more workers**.
- When the price goes down, the firm hires **fewer workers**.

2. Prices of Other Inputs (like machines)

- If machines become cheaper, firms may replace workers with machines, so **labor demand falls**.
- Sometimes cheaper machines allow firms to produce more, which can lead to **more workers being hired**.

3. Technology

- New technology can replace some jobs, reducing demand for certain workers.
- At the same time, it creates new jobs, increasing demand for other types of workers.

Changes in Demand

(Shifts in the demand curve for labor)

A firm's demand for labor

Decreases if:

- The price of the firm's output decreases
- The price of a substitute for labor falls
- The price of a complement of labor rises
- A new technology or new capital decreases the marginal product of labor

Increases if:

- The price of the firm's output increases
- The price of a substitute for labor rises
- The price of a complement of labor falls
- A new technology or new capital increases the marginal product of labor

The Market Supply of Labor

The market supply of labor is derived from the supply of labor decisions made by individual households.

Individual's Labor Supply Decision

- People divide their time between **working** and **leisure** (everything that is not work).
- Most people prefer leisure to work, so they need to be **paid a wage** to choose to work.

Reservation Wage

- The **reservation wage** is the **minimum wage** a person is willing to accept to work.
- If the wage is **below** this level, the person chooses leisure instead of working.
- If the wage is **equal to or above** the reservation wage, the person is willing to **supply labor**.

Example:

- Sonny values his free time, but he is willing to work if he earns **at least \$5 per hour**.
- At wages above \$5 per hour, Sonny chooses to work.

Backward-Bending Labor Supply Curve

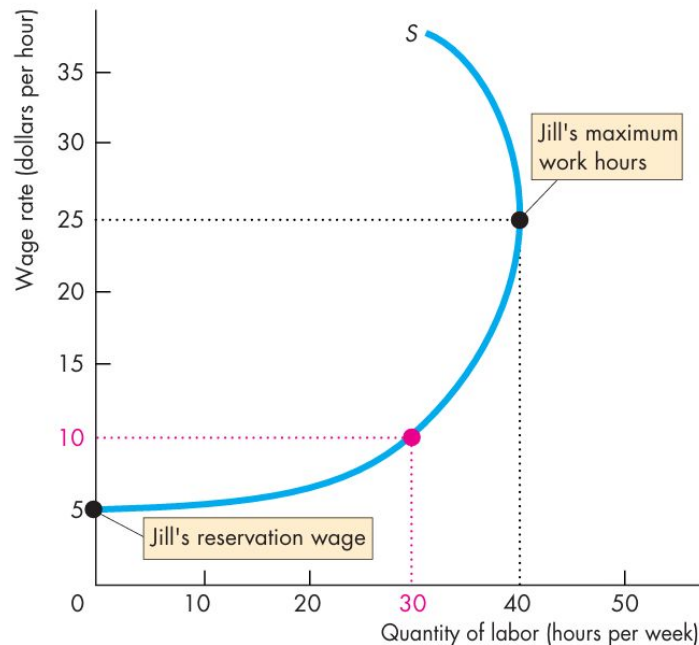
Jill's Labor Supply Curve

- Jill's labor supply curve shows how many hours she is willing to work at different wages.
- **Below \$5/hour** → Jill works **0 hours** (her reservation wage).
- **\$5–\$25/hour** → As the wage rises, Jill works **more hours**, up to 40 hours per week.
- **Above \$25/hour** → Jill works **fewer hours** even as wages rise. This is called a **backward-bending labor supply curve**.

Why it bends backward:

- At very high wages, Jill values **leisure more than extra income**, so the **income effect** dominates the **substitution effect**.

FIGURE 18.2 Jill's Labor Supply Curve



1. Substitution Effect

- Jill's **wage is the cost of leisure**.
- If she quits work early, she gives up income equal to her wage.
- **Higher wages** make giving up work more expensive, so Jill is more likely to **work longer hours**.

2. Income Effect

- Higher wages mean **higher income**.
- With more income, Jill can afford **more leisure**.
- As a result, she may **work fewer hours** despite the higher wage.

Equilibrium in a Competitive Labor Market

- A competitive labor market **matches firms' demand for workers with households' supply of labor**.
- **Equilibrium** occurs when the **quantity of labor demanded equals the quantity supplied**.
- *Example:* At a wage of **\$10/hour**, **300 workers** are employed, and the market is in balance.

If wages are too high:

- More people want to work than firms want to hire.
- This **excess supply** causes wages to **fall**.

If wages are too low:

- Firms want to hire more workers than are willing to work.
- This **excess demand** causes wages to **rise**.

FIGURE 18.3 The Market for Bakery Workers

